VS-02 Data Standardization and Integration

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VS-02 Data Standardization and Integration

1. Purpose and value created

VS-02 makes data from different sources speak the same language. Once raw data is in NSI custody, VS-02 aligns its structure, format, and codes with the agreed standards of the NSI, so that datasets that looked different at the source can be compared and combined. It does this without changing any value: it re-expresses and annotates, it does not transform. The result is the same information in a common shape, expressed in standard classifications, with rich metadata and without direct identifiers. The value created is **coherence and comparability**: data from surveys, administrative registers, and external sources can flow into editing, analysis, and modelling as one structured, standardised asset.

VS-02 corresponds to the business domain **Standardization without loss of content** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management. Content correction, imputation, and enrichment are out of scope; they belong to VS-03 Data Editing and Enrichment.

> **i** Signature discipline: standardisation without loss of content
>
> VS-02 only maps codes one-to-one, reformats structure, and annotates metadata. It never changes a value. Unit and currency conversion, reference-period alignment, value reconciliation across sources, derivation of new variables, and interpretive coding of free text are deliberately left to VS-03. Interpretive coding, for example coding a written job title to ISCO, needs judgement that can change informational content, so it is not a VS-02 activity. The no-loss-of-content rule covers the statistical content: records, observed values, and key measures. It does not cover direct identifiers, which are personal data rather than statistical content. Their removal or pseudonymisation at this transition (SAF-P11) therefore never counts as content loss.

PLACEHOLDER: ArchiMate view VS-02 (a) value stream and capabilities

2. Position and state transition

VS-02 is the second primary value stream. It receives raw data from VS-01 and hands standardised data to VS-03. It is the stage that harmonises and integrates information, so that data from different sources works together, by aligning structure, format, and codes and not by changing values.

VS-02 takes data from State #1 to State #2. Before VS-02, the data is in NSI custody in its original form, usually with direct identifiers. After VS-02, it is aligned to common structures, formats, classifications, and semantics, with direct identifiers removed or pseudonymised, so that it is interoperable and comparable across sources, domains, and time. The quality gate, the minimum metadata set, and the protection of State #2 are described in *State #2: Standardized Data* and are not repeated here. **[VC10]** State #2 is optional: a source that already arrives in the standard structure and coding of the NSI can go directly from #1 to VS-03, and the skip is recorded in the catalogue.

From	To	Gate	Event out
#1 Raw Data	#2 Standardized Data	Gate of State #2 (structural and classification conformance, mapping coverage, semantic metadata, no loss of content, identifier protection, referential integrity, metadata and lineage)	*Standardized Data Registered*

PLACEHOLDER: ArchiMate view VS-02 (c) state handover and events: Raw Data (#1) from VS-01, the standardisation service at the domain boundary (profiling, classification and coding, semantic alignment, conformance check), Standardized Data (#2) handed to VS-03

3. Stages

The stages describe the value that VS-02 adds in every production cycle. Together they form the execution process type of VS-02. In a mature steady state, the target model, correspondence tables, and conformance rules have been designed and built in advance (see attribute 4). Execution then focuses on applying mappings and classifications, capturing lineage and conformance metadata, and passing the quality gate without altering content.

Stage	Name	Value added
VS-02.1	Profile and Assess Data Structure	Differences from the target model are known
VS-02.2	Apply Standard Classifications and Formats	Data uses common codes, formats, and identifiers
VS-02.3	Align Metadata and Semantics	Definitions and descriptions are consistent
VS-02.4	Validate Structural and Semantic Conformance	Conformance to standards is confirmed
VS-02.5	Register Standardized Artefacts	Standardised data is registered and available for reuse (State #2)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies. **[VC9]**

VS-02.1 Profile and Assess Data Structure

Value created: *A clear understanding of how far incoming raw data is from the target standard, so the right mapping and format coercion can be applied.*

The stage starts when the *Raw Data Registered* event arrives from VS-01. The profiling components analyse the structure of the incoming dataset: they infer types, measure cardinalities and value distributions, and detect inconsistencies, missing fields, and structural anomalies relative to the target data model. The dataset is then compared with the target schema. The result is a structure-gap assessment: which fields map directly, which need format coercion, which source codes have a standard correspondence, and which values are uncoded and must go to VS-03. A significant or unexpected structural deviation, such as a source that has changed its schema, goes back to Design instead of being adapted ad hoc during the cycle.

Activity	Key outputs	SAF process	Other reference
Profile incoming raw data: types, cardinalities, distributions, anomalies	Data profiling report	PRD-5.3 Review and validate	GSBPM 5.1
Compare the structure with the target data model	Structure-gap assessment	PRD-5.3 Review and validate	
Identify the format coercion and code mappings to apply	Mapping plan (per dataset)	PRD-5.3 Review and validate; PRD-5.2 Classify and code	
Flag unexpected structural deviations for design review	Schema-change exception (if any)	PRD-5.3 Review and validate	

VS-02.2 Apply Standard Classifications and Formats

Value created: *Source codes re-expressed in shared, authoritative classifications and a common format, so data is comparable across sources, with no value changed.*

This is the core of standardisation. The mapping and format-coercion component applies the designed value-preserving operations to bring the data into the target format: renaming, restructuring, type casting, reformatting (for example from CSV to XML), and key alignment. The mapping services then translate values that already carry a source code into the corresponding standard code (NACE, ISCO, COICOP, NUTS, SDMX code lists) through deterministic correspondence tables, one-to-one, keeping the original value. Values without a correspondence, and free text that needs interpretation, are not coded here. They are flagged and routed to VS-03. Only the representation changes and standard codes are added. Every applied mapping and classification version is recorded as lineage.

Activity	Key outputs	SAF process	Other reference
Apply format coercion: rename, restructure, cast, reformat, align keys	Format-coerced dataset (values unchanged)	PRD-5.1 Integrate data (key alignment only)	GSBPM 5.1
Map source codes to standard classifications one-to-one through correspondence tables	Standard-coded dataset (original preserved)	PRD-5.2 Classify and code; DLM-9 Manage reference and master data	1.3.1 Data standardisation
Flag unmapped, uncoded, and free-text values and route them to VS-03	Unmapped-value handover list	PRD-5.2 Classify and code	

Activity	Key outputs	SAF process	Other reference
Capture applied mappings and classification versions as lineage	Mapping and classification lineage	PRD-5.2 Classify and code; OVA-2 Metadata Management	SAF-P09

PLACEHOLDER: VS-02 classification mapping logic: survey, administrative, and emerging-source inputs pass format coercion and a deterministic correspondence lookup; values whose source code has a correspondence become standard-coded values with the original preserved, unmapped and free-text values go to VS-03 for interpretive coding

VS-02.3 Align Metadata and Semantics

Value created: *Coherent, shared meaning across datasets, so the same concept means the same thing everywhere.*

Once structure and codes are standardised, descriptive metadata and semantics are harmonised at the metadata level only. Source variables are linked to canonical concepts in the concept system. The unit of measure and reference-period basis of each variable are recorded as metadata against the standard conventions, not converted. Where sources define the same concept differently, for example two sources that define "employee" differently, the difference is recorded as metadata and flagged for VS-03 instead of being reconciled by adjusting values. Quality descriptors (completeness, accuracy, confidentiality flags) are standardised and attached. The harmonised metadata is bound to the dataset, so downstream consumers receive both the standardised data and the semantics needed to interpret it.

Activity	Key outputs	SAF process	Other reference
Link source variables to canonical concepts	Concept-aligned variables	OVA-2 Metadata Management	GSBPM 5.1
Record unit of measure and reference-period basis as metadata (no conversion)	Unit and period metadata	OVA-2 Metadata Management	SAF-P09
Record differing cross-source definitions as metadata and flag them for VS-03	Definition-difference record	OVA-2 Metadata Management	2.1.4 Statistical register management
Standardise and attach quality descriptors	Harmonised quality metadata	OVA-2 Metadata Management	CSDA Metadata Management

VS-02.4 Validate Structural and Semantic Conformance

Value created: *Confidence that the standardised data conforms to standards and that no content was lost in standardisation.*

Before standardised data can be accepted, it must pass the quality gate of State #2. The automated checks from *Design Conformance Validation Rules* run on every dataset: structural validation against the target model, classification conformance, and semantic conformance. Content-preservation checks confirm that every value is unchanged and that record counts and key business measures are identical before and after. The conformance assessment gives a structural score, a mapping coverage rate, and a semantic conformance rating. Every result, exception, and decision is logged in a mapping and format-coercion audit trail, with data metadata (what was validated, what passed) and process metadata (who validated, when, which rule and engine version). The outcome is one of two business events. *Standardisation Conformed* starts registration. *Standardisation Non-Conformant* starts exception handling: re-map, route unmapped values to VS-03, or accept conditionally with a conformance flag.

Activity	Key outputs	SAF process	Other reference
Validate the structure against the target model	Structural validation result	PRD-5.3 Review and validate; DLM-2 Promote data through a quality gate	
Validate classification conformance: valid codes, correct version, coverage	Classification conformance result	PRD-5.3 Review and validate	
Validate semantic conformance: units, definitions, reference periods recorded	Semantic conformance result	PRD-5.3 Review and validate	
Run the no-loss-of-content checks: record and measure preservation	Content-preservation result	DLM-2 Promote data through a quality gate	GAMSO Quality Management
Create the audit trail: log all conformance results, exceptions, and decisions	Audit trail record	OVA-4 Process Data Management	GAMSO Quality Management

VS-02.5 Register Standardized Artefacts

Value created: *Secure, organised, and traceable storage of standardised data, discoverable by downstream consumers.*

This stage persists conformant standardised data and makes it discoverable for the rest of the statistical system. Each dataset version receives a unique persistent identifier (URI or UUID) and is stored in the standardised data repository, in line with the loose coupling of SAF-P07. The stage confirms that identifiers have been removed or pseudonymised according to the privacy design (SAF-P11). Metadata is registered in the catalogue, in line with SAF-P09: the mapping lineage, the applied classification versions, the harmonised concept and quality metadata, and the conformance results. Lineage from Raw Data (#1) to Standardized Data (#2) is recorded, traceable to the originating source. Access permissions are applied according to the designed framework (SAF-P10). The stage ends by publishing the *Standardized Data Registered* event, which tells VS-03 that new standardised data is available for editing and enrichment.

Output: Standardized Data (State #2), the official handover artefact to the rest of the statistical value chain. Its structure and minimum metadata set are described in *State #2: Standardized Data.* **[VC10]**

Activity	Key outputs	SAF process	Other reference
Assign a unique persistent identifier (URI or UUID) to the dataset version	Dataset ID	DLM-3 Register and catalogue data	
Store the data in the standardised data repository	Stored standardised artefact	DLM-5 Store and version data	SAF-P07
Confirm that identifier removal or pseudonymisation has been applied	Privacy-compliant dataset	DLM-4 Classify and protect data	SAF-P11
Register metadata: mapping lineage, classification versions, harmonised concepts, conformance results	Catalogue registration	DLM-3 Register and catalogue data; OVA-2 Metadata Management	SAF-P09

Activity	Key outputs	SAF process	Other reference
Apply access permissions according to the designed framework	Access-controlled dataset	DLM-4 Classify and protect data	SAF-P10
Publish the *Standardized Data Registered* event to notify VS-03	Event notification	DLM-3 Register and catalogue data	SAF-P07

4. The four process types

The sub-processes below are specific to VS-02. Each refines one or more SAF business processes, listed as *Refines*, which keeps it traceable to *Business Processes*. Sub-processes marked *(standard)* work as described in Standard activities in every value stream; only the VS-02 specifics are listed. In this domain, metadata, and classification in particular, is not a by-product but the central asset being created.

Design: provides metadata

Design sub-process	Refines	Main outputs
Design Target Data Models and Schemas	PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-1 Plan and design data assets; DLM-9 Manage reference and master data	Target data model, canonical concept catalogue, structural conformance rules, integration key design
Design Classification Mapping Strategy	PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	Classification standards register, version policy, correspondence tables, unmapped-value routing rules
Design Metadata and Semantic Harmonisation Rules	PRD-2.2 Design variables; OVA-2 Metadata Management	Harmonised concept system, unit and reference-period metadata standards, semantic mapping rules, definition-difference policy
Design Conformance Validation Rules	PRD-2.5 Design processing and analysis; DLM-2 Promote data through a quality gate; OVA-1 Quality Management	Validation rule sets, content-preservation checks, exception handling policy
Design Standardisation Registration and Privacy Framework	DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-4 Process Data Management	Data classification policy, access control design, lineage model, pseudonymisation design, audit logging specification

Design Target Data Models and Schemas

Refines: PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-1 Plan and design data assets; DLM-9 Manage reference and master data.

Standardisation starts from the canonical structure that all data must conform to. This sub-process specifies the target data model for Standardized Data (State #2): entities, fields, data types, formats, units of measure, cardinalities, and referential constraints. It sets canonical representations for common concepts, such as dates, geographies, currencies, and organisational and personal identifiers, so that data from surveys, registers, and external sources can be expressed consistently. It defines the structural conformance contract: what it means for a dataset to be structurally valid against the target model. Where several sources describe the same population, the integration model defines how they are aligned to a common key structure, for example the key of a business or statistical unit register.

Activity	Key outputs	SAF process	Other reference
Define the canonical target data model for State #2: entities, fields, types, formats, units	Target data model specification	PRD-2.2 Design variables; DLM-1 Plan and design data assets	1.1.2 Design statistics product
Define canonical representations for common concepts (dates, geography, currency, identifiers)	Canonical concept catalogue	PRD-2.2 Design variables	
Define the structural conformance contract and referential constraints	Structural conformance rules	PRD-2.5 Design processing and analysis	
Define the integration and linkage key structure for multi-source alignment	Integration key design	DLM-9 Manage reference and master data	2.1.4 Statistical register management

Design Classification Mapping Strategy

Refines: PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data.

Once the target structure is set, the classification steward designs how source codes are mapped one-to-one to standard classifications. The sub-process selects the authoritative classifications and code lists the NSI conforms to, such as NACE, ISCO, COICOP, NUTS, SDMX code lists, and concepts aligned with GSIM. It defines the version policy: which version applies, and how transitions between versions are handled. The mapping is deterministic: maintained correspondence tables translate a value that already carries a source code into the corresponding standard code, and the original value is kept. The design also specifies that unmapped or uncoded values are flagged and passed to VS-03, which does any interpretive coding of free text. Reference and master data sources are identified, so that mapping is anchored to authoritative registers.

Activity	Key outputs	SAF process	Other reference
Select authoritative classifications and code lists (NACE, ISCO, COICOP, NUTS, SDMX, GSIM concepts)	Classification standards register	PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	
Define the classification version policy and transition handling	Classification version policy	DLM-9 Manage reference and master data	SAF-P09
Design deterministic one-to-one correspondence tables from source codes to standard codes	Correspondence table set	PRD-2.5 Design processing and analysis	1.3.1 Data standardisation
Define the handling of unmapped and uncoded values: flag and route to VS-03	Unmapped-value routing rules	PRD-2.5 Design processing and analysis	
Identify reference and master data sources for mapping	Reference data catalogue	DLM-9 Manage reference and master data	CSDA Reference Data

Design Metadata and Semantic Harmonisation Rules

Refines: PRD-2.2 Design variables; OVA-2 Metadata Management.

This sub-process defines how descriptive metadata and semantics are harmonised, so that data from different sources is understood consistently without any value changing. The team designs harmonised concept and variable definitions, and the metadata that records the unit of measure and reference-period basis of each variable. Semantic mapping rules link source variables to canonical concepts in the concept system, and quality descriptors (completeness, accuracy, confidentiality flags) are standardised. Where sources define the same concept differently, the design specifies that the difference is recorded as metadata and flagged for VS-03; VS-02 does not adjust values to reconcile them. The catalogue structure for standardised artefacts is designed so that lineage from raw to standardised state is fully traceable, which makes active metadata a foundation for downstream automation, as emphasised by the Metadata Management capability of CSDA.

Activity	Key outputs	SAF process	Other reference
Define harmonised concept and variable definitions	Harmonised concept system	PRD-2.2 Design variables	
Define the unit and reference-period metadata to record per variable (no conversion)	Unit and reference-period metadata standards	OVA-2 Metadata Management	SAF-P09
Design semantic mapping rules from source variables to canonical concepts	Semantic mapping rules	OVA-2 Metadata Management	CSDA Metadata Management
Design rules to record differing cross-source definitions as metadata and flag them for VS-03	Definition-difference policy	OVA-2 Metadata Management	2.1.4 Statistical register management
Design the catalogue structure and lineage model for standardised artefacts	Catalogue registration template	OVA-2 Metadata Management	CSDA Metadata Management

Design Conformance Validation Rules

Refines: PRD-2.5 Design processing and analysis; DLM-2 Promote data through a quality gate; OVA-1 Quality Management.

Before any data is standardised, the criteria for "standardised" must be clear. This sub-process sets structural validation rules (field types, formats, value ranges, cardinalities, referential integrity against the target model), classification conformance rules (every mapped code belongs to the correct classification version, no orphan codes), and semantic conformance rules (units, definitions, and reference periods correctly recorded as metadata). The maximum acceptable unmapped-value rate is agreed as a coverage threshold. Exception handling rules distinguish between rejection, conditional acceptance with conformance flags, and routing of unmapped values to VS-03. The rules also verify the no-loss-of-content rule: every value is unchanged, and record counts and key business measures are identical before and after. Together, these rules make up the VS-02 part of the quality gate of State #2. The Quality Management capability of GAMSO provides the overall control framework.

Activity	Key outputs	SAF process	Other reference
Define structural validation rules against the target model	Structural validation rule set	PRD-2.5 Design processing and analysis	
Define classification conformance rules: valid codes, correct version, no orphans	Classification conformance rules	PRD-2.5 Design processing and analysis	

Activity	Key outputs	SAF process	Other reference
Define semantic conformance rules: units, definitions, reference periods	Semantic conformance rules	PRD-2.5 Design processing and analysis	
Define the no-loss-of-content checks: record and measure preservation	Content-preservation checks	OVA-1 Quality Management	GAMSO Quality Management
Design exception handling rules: rejection, conditional acceptance, manual review	Exception handling policy	DLM-2 Promote data through a quality gate	

Design Standardisation Registration and Privacy Framework

Refines: DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-4 Process Data Management.

This sub-process makes sure that standardised artefacts are correctly registered, traceable, and privacy-compliant. Each standardised dataset receives a classification for availability, integrity, and confidentiality (SAF-P09), and role-based access control for the standardised data repository is designed (SAF-P10). The lineage model is designed so that every standardised field can be traced back to its raw source value and the mapping rule applied. Privacy protection follows SAF-P11: standardisation is where direct identifiers are removed or replaced with pseudonymous keys. The design specifies which identifiers are dropped, which are pseudonymised, the pseudonymisation method and key management, and the data minimisation rules. Audit logging requirements, which define the mapping and format-coercion events that must be recorded, complete the design.

Activity	Key outputs	SAF process	Other reference
Define the data classification (availability, integrity, confidentiality) for standardised datasets	Data classification policy	DLM-4 Classify and protect data	SAF-P09
Design role-based access control for the standardised data repository	Access control design	DLM-1 Plan and design data assets; DLM-4 Classify and protect data	SAF-P10; 3.3.4 Logical identity and access control
Design the lineage model: standardised field, raw source value, mapping rule	Lineage model specification	DLM-1 Plan and design data assets	CSDA Traceability
Design identifier removal and pseudonymisation: which identifiers, method, key management, minimisation	Privacy and pseudonymisation design	DLM-4 Classify and protect data	SAF-P11; 3.3.1 Privacy management
Design audit logging requirements for mapping and format coercion	Audit logging specification	OVA-4 Process Data Management	SAF-P10

PLACEHOLDER: VS-02 design process decomposition: inputs (raw data model of State #1, framework standards, classification standards such as NACE, ISCO, COICOP, SDMX, legal and privacy requirements), the five design sub-processes in order, and the artefact each produces (target data model, correspondence tables, harmonised concepts, conformance rule set, lineage and privacy design), which together govern Implementation, Execution, and Management

Implementation: provides the execution process

Implementation sub-process	Refines	Main outputs
Build Profiling and Mapping Engines	PRD-3.2 Reuse or build processing and analysis components	Profiling module, lineage-aware mapping and format-coercion component, versioned mapping repository
Configure Classification Mapping Services	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data	Mapping services with loaded correspondence tables, version control, unmapped-value routing
Build Conformance Validation Pipelines and Metadata Registration	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-3 Register and catalogue data; DLM-4 Classify and protect data; OVA-2 Metadata Management	Automated conformance pipeline, pseudonymisation component, metadata propagation, events, exception routing
Test and Validate the Pipeline *(standard)*	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalise production systems	Test reports; mapping verification report (correspondence tables checked against the source code lists); pilot confirming one-to-one mapping, complete lineage, no loss of content, routing of unmapped values to VS-03, and identifier removal; volumes tested against processing windows

Build Profiling and Mapping Engines

Refines: PRD-3.2 Reuse or build processing and analysis components.

This sub-process turns the target data model and mapping designs into operational tools. Profiling components analyse the structure of incoming raw data: they infer types and detect cardinalities, value distributions, and structural anomalies, so the pipeline can report how far each source is from the target model. The mapping and format-coercion component applies the designed value-preserving operations: renaming, restructuring, type casting, reformatting (for example from CSV to XML), and key alignment. It does not convert units, currencies, or reference periods; those are recorded as metadata, and any conversion takes place in VS-03. The component is declarative and lineage-aware: for each output field it records which input field and which rule produced it. All mapping logic is version-controlled, so that any standardised record can be reproduced from raw data.

Activity	Key outputs	SAF process	Other reference
Build data profiling components: type inference, cardinality, distributions, anomalies	Production-ready profiling module	PRD-3.2 Reuse or build processing and analysis components	3.2.1 Application development

Activity	Key outputs	SAF process	Other reference
Build or configure the mapping and format-coercion component: rename, restructure, cast, reformat, align keys	Configured mapping component	PRD-3.2 Reuse or build processing and analysis components	
Configure lineage capture for every mapping and coercion operation	Lineage-aware mapping modules	PRD-3.2 Reuse or build processing and analysis components	CSDA Traceability
Version-control all mapping and format-coercion logic	Versioned mapping repository	PRD-3.2 Reuse or build processing and analysis components	SAF-P10

Configure Classification Mapping Services

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data.

The designed correspondence tables are turned into automated mapping services. Deterministic correspondence tables are loaded and connected to authoritative reference data, so that a value already carrying a source code is translated one-to-one to the corresponding standard code. Classification version control is configured, so that the correct version is applied and version transitions are managed. Values without a matching correspondence, and free-text or uncoded values, are routed to VS-03 instead of being coded here. Reference and master data connections keep the mapping synchronised with authoritative registers.

Activity	Key outputs	SAF process	Other reference
Load correspondence tables for deterministic mapping from source codes to standard codes	Configured mapping services	PRD-3.2 Reuse or build processing and analysis components	
Configure classification version control and transition handling	Version-controlled classifications	DLM-9 Manage reference and master data	SAF-P09
Configure the routing of unmapped and uncoded values to VS-03	Unmapped-value routing	PRD-3.4 Configure workflows	
Configure reference and master data connections	Synchronised reference data	DLM-9 Manage reference and master data	CSDA Reference Data

Build Conformance Validation Pipelines and Metadata Registration

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-3 Register and catalogue data; DLM-4 Classify and protect data; OVA-2 Metadata Management.

This sub-process implements the conformance validation rules and the registration of standardised artefacts. Structural, classification, semantic, and content-preservation checks run as automated validations on every standardised dataset. Catalogue entries are built from the designed metadata model, and automatic propagation makes mapping lineage, applied classification versions, harmonised definitions, and conformance scores flow into the catalogue. Identifier removal and pseudonymisation are implemented according to the privacy design. Event notifications alert VS-03 when standardised data is registered, and exception workflows route non-conforming data to manual review. As CSDA emphasises, the ultimate goal is active metadata: metadata that can drive standardisation rules and downstream processing specifications.

Activity	Key outputs	SAF process	Other reference
Implement conformance checks as automated validations (structural, classification, semantic, content preservation)	Automated conformance pipeline	PRD-3.2 Reuse or build processing and analysis components	
Implement identifier removal and pseudonymisation according to the privacy design	Pseudonymisation component	DLM-4 Classify and protect data	SAF-P11
Configure automatic metadata and lineage propagation into the catalogue	Automated metadata pipelines	DLM-3 Register and catalogue data; OVA-2 Metadata Management	CSDA Metadata Management
Configure the *Standardized Data Registered* event for downstream consumers	Event publication	PRD-3.4 Configure workflows	SAF-P07
Configure exception handling workflows	Non-conformance routing and review queues	PRD-3.4 Configure workflows	

PLACEHOLDER: VS-02 implementation pipeline: the five design artefacts feeding the three build sub-processes (target model and harmonisation rules into the profiling and mapping engines, the classification strategy into the mapping services, conformance rules and the lineage and privacy design into the conformance pipeline), followed by technical testing, pilot and mapping verification, and performance and lineage testing, resulting in a production-ready standardisation pipeline and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-02 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts for each new raw data version registered by VS-01. Execution applies the designs and the solution as they are; a changed source schema or a missing correspondence that needs a new rule goes back to Design. Every dataset ends in one of three outcomes: registered, registered with conformance flags and a list of unmapped values for VS-03, or returned for re-mapping.

Aspect	VS-02
Trigger	*Raw Data Registered*
Cadence	Per registered raw data version
Performed by	ROL-14 Data quality analyst, ROL-03 Data steward
Input	Raw data version (#1), production-ready pipeline, current correspondence tables and target model
Output	Registered standardised data version (#2), its mapping lineage, and the list of unmapped values for VS-03
Control point	Gate of State #2 in VS-02.4
Exception path	Re-mapping after *Standardisation Non-Conformant*; unmapped values routed to VS-03 with conditional acceptance; schema change at the source back to Design

Management: provides quality reports

Management sub-process	Refines	Main outputs
Monitor Process Performance *(standard)*	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard; throughput report against processing windows; paradata on engine and classification versions
Assess Standardisation Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Mapping completeness report, conformance and orphan-code report, conformance quality report, trend analysis
Manage Classification and Metadata Lifecycle	DLM-9 Manage reference and master data; OVA-2 Metadata Management; DLM-8 Monitor data quality, lineage and use	Classification version and correspondence register, metadata quality report, reference data maintenance log, lineage audit
Audit and Compliance Reporting *(standard)*	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Audit trail from each raw value through the mapping rules to the standardised value, supporting reproducibility and content preservation; privacy compliance record on identifier removal, pseudonymisation, and minimisation at the #1 to #2 transition; standards compliance report on adopted classifications and SDMX exchange (4.1.5 Accountability)
Continuous Improvement *(standard)*	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, for example extending correspondence tables to raise coverage, refining mappings that produce orphan codes, adopting new classification versions, and simplifying conformance rules that produce false positives; recurring unmapped values that need interpretive coding are passed to VS-03 instead of being absorbed in VS-02; trends watched: mapping coverage, unmapped-value rate, conformance pass rate, re-mapping rate

Assess Standardisation Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality of the standardised data itself is evaluated against the criteria set in *Design Conformance Validation Rules*. Each dataset is assessed on several dimensions. Mapping completeness covers the share of source codes mapped one-to-one and the unmapped-value rate routed to VS-03. Classification conformance covers orphan codes and correct versions. Structural and format conformance covers alignment with the target model. Metadata completeness covers units, reference-period bases, and definitions correctly recorded. Content preservation confirms that no value changed and that record counts and key measures are identical. Quality indicators are produced per dataset, per source, and per production cycle. Trends are tracked to detect deteriorating sources, growing unmapped-value rates, or

problems with classification versions. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Assess mapping completeness and the unmapped-value rate	Mapping completeness report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess classification conformance and the orphan-code rate	Conformance and orphan-code report	OVA-1 Quality Management	CSDA Data Quality
Assess structural and format conformance, metadata completeness, and content preservation	Conformance quality report	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management
Identify quality trends across sources, classifications, and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management

Manage Classification and Metadata Lifecycle

Refines: DLM-9 Manage reference and master data; OVA-2 Metadata Management; DLM-8 Monitor data quality, lineage and use.

The classifications and metadata that support standardisation must stay complete, accurate, and current. This sub-process governs the life cycle of classification versions: adopting new versions, for example a NACE or COICOP revision, managing transition periods, and maintaining the correspondence tables between versions. It verifies, as a compliance check on SAF-P09, that every standardised dataset carries full mapping lineage, classification versions, and harmonised concepts. It maintains the concept system, code lists, and reference data, so that mapping stays anchored to authoritative sources. Lineage from raw through standardisation to the standardised state is tracked, and reconciliation across systems makes sure that catalogue entries match the stored artefacts. As CSDA emphasises, active, well-managed metadata is what makes standardised data reusable and downstream automation possible.

Activity	Key outputs	SAF process	Other reference
Govern the classification version life cycle and correspondence tables	Classification version and correspondence register	DLM-9 Manage reference and master data	SAF-P09
Verify metadata completeness for all standardised datasets	Metadata quality report	OVA-2 Metadata Management	SAF-P09
Maintain the concept system, code lists, and reference data	Reference data maintenance log	DLM-9 Manage reference and master data; OVA-2 Metadata Management	CSDA Metadata Management
Track metadata lineage from raw through standardisation to the standardised state	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Traceability
Reconcile metadata across systems	Catalogue-artefact reconciliation	OVA-2 Metadata Management	SAF-P09

PLACEHOLDER: VS-02 management feedback loop: data and paradata from Execution flowing through monitoring (key performance indicators, throughput), standardisation quality (mapping coverage), classification and metadata lifecycle (versions, reference data), audit (lineage, privacy), and continuous improvement, with dashboards to the process owner, quality reports to the quality manager, compliance reports to VS-07, and updated rules and classifications back to Design

Metadata per process type

Metadata is produced in every process type, not only in Design. The table shows the data metadata each process type of VS-02 contributes; the paradata are listed in Standard activities in every value stream. In Execution, VS-02 also records the mapping coverage, the count of unmatched values, and the engine and classification versions as paradata, and Design keeps a classification version policy.

Process type	Data metadata produced
Design	Target data models, classification schemes, code lists, source-to-target mapping rules, conformance criteria, harmonised definitions
Implementation	Mapping configurations, parameters of coding services and validation rules, catalogue templates
Execution	Per dataset: source-to-target mappings, applied classification versions, mapping and format-coercion lineage, conformance scores, profiling statistics
Management	Conformance scores, classification quality assessments, coverage reports, lineage assessments, trend analyses

PLACEHOLDER: VS-02 business object lifecycle: design artefacts (target model, classification strategy, harmonisation, conformance, and privacy design) govern implementation assets (mapping and format coercion, mapping services, conformance pipeline, catalogue configuration), which produce the process that standardises Raw Data (#1) into Standardized Data (#2), generating mapping lineage, conformance, and audit records that management assesses and feeds back to design

5. Actors and roles

VS-02 works mostly inside the NSI. Its upstream partner is VS-01, which supplies Raw Data (#1) and its provenance metadata through the *Raw Data Registered* event. The table shows the actors and the reference roles involved. Specialised role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the data architect and the metadata architect of the original VS-02 description are both ROL-08 Metadata and data architect, and the system configurator is ROL-11 Platform engineer.

Actor or role	ID	Process type	Responsibility in VS-02
International statistical organisation (standards body)	ACT-07	External	Maintains the classification and exchange standards the NSI conforms to, for example Eurostat and UNSD for NACE, ISCO, COICOP, and SDMX
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-02

Actor or role	ID	Process type	Responsibility in VS-02
Data owner	ROL-02	Management	Accountable for State #2: conformance and the application of the privacy design
Data steward	ROL-03	Execution	Registers standardised datasets, manages metadata entries, and applies access controls
Data custodian	ROL-04	Execution	Operates storage, access control, and retention; manages pseudonymisation keys separately from the data
Methodologist	ROL-07	Design	Designs the mapping methodology and the cross-source harmonisation of definitions and metadata
Metadata and data architect	ROL-08	Design	Designs the target data models, canonical schemas, structural conformance rules, harmonised definitions, semantic mappings, unit and period metadata, and the standardised metadata schema
Classification steward	ROL-09	Design	Owns the classification strategy, code lists, and the one-to-one correspondence tables from source codes to standards
Data engineer	ROL-10	Implementation	Builds and configures the profiling, mapping, and format-coercion components and the standardisation pipeline
Platform engineer	ROL-11	Implementation	Configures classification services, reference data connections, and access controls
Test manager	ROL-12	Implementation	Plans and carries out testing of the standardisation pipeline: technical, conformance, and integration tests
Data quality analyst	ROL-14	Execution	Profiles data, assesses structural and semantic conformance, and logs unmapped values and exceptions

Actor or role	ID	Process type	Responsibility in VS-02
Quality manager	ROL-18	Management	Monitors conformance key performance indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures privacy (identifier removal) and standards compliance, and produces audit reports

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-02 relies on, and what role they play. NSIs can use it to map the value stream to their own organisation.

Capability group	Capabilities	Role in VS-02
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the target data models and classification strategies, and build the standardisation pipeline
1.3 Processing	1.3.1 Data standardisation	Standardise formats, category codes, linkage keys, and the recording of units without loss of content, directly after receipt of raw data
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage data and metadata through standardisation, harmonise semantics, assess conformance quality, align with reference registers
3.2 Computerisation and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain mapping engines, coding services, and conformance validation pipelines
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.4 Logical identity and access control	Remove or pseudonymise direct identifiers and build security into the standardisation processes
4.1 Governance	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, steward classification standards, plan improvements, report on compliance

Stages, processes, and functions

The table shows which SAF business processes realise each stage of VS-02 and which business functions they mainly use.

Stage	Realised by processes	Main functions
VS-02.1 Profile and Assess Data Structure	PRD-5.3 Review and validate	FUN-3.3 Process statistical data
VS-02.2 Apply Standard Classifications and Formats	PRD-5.2 Classify and code; DLM-9 Manage reference and master data	FUN-3.3 Process statistical data; FUN-7.4 Manage reference and master data

Stage	Realised by processes	Main functions
VS-02.3 Align Metadata and Semantics	OVA-2 Metadata Management	FUN-7.2 Manage metadata
VS-02.4 Validate Structural and Semantic Conformance	PRD-5.3 Review and validate; DLM-2 Promote data through a quality gate	FUN-3.3 Process statistical data; FUN-7.3 Manage data lifecycle and storage
VS-02.5 Register Standardized Artefacts	DLM-3 Register and catalogue data; DLM-4 Classify and protect data; DLM-5 Store and version data	FUN-7.2 Manage metadata; FUN-7.3 Manage data lifecycle and storage; FUN-7.5 Protect data

PLACEHOLDER: ArchiMate view VS-02 (b) stages realised by processes and functions

Framework mappings

This reference table maps each process type of VS-02 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-2.2, PRD-2.5, DLM-1, DLM-2, DLM-4, DLM-9, OVA-1, OVA-2, OVA-4	Phase 2 Design (2.2, 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management, Reference and Master Data	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.4 Statistical register management
Implementation	PRD-3.2, PRD-3.4 to PRD-3.7, DLM-3, DLM-4, DLM-9, OVA-2	Phase 3 Build (3.2, 3.4 to 3.7)	Capability Management	Data Integration (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-5.1 (key alignment only), PRD-5.2, PRD-5.3, DLM-2 to DLM-5, DLM-9, OVA-2, OVA-4	Phase 5 Process (5.1 Integrate data, key alignment only; 5.2 Classify and code; 5.3 Review and validate, structural checks only)	Production (execution)	Data Integration (execution)	1.3.1 Data standardisation, 2.1.2 Metadata management, 2.1.4 Statistical register management

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Management	OVA-1, OVA-2, OVA-4, DLM-8, DLM-9, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-02 most directly.

Principle	Statement	Application in VS-02
SAF-P01	Actively participate in open-source communities to maintain collective control over statistical processes	Classification, coding, and mapping tooling, such as SDMX libraries or coding engines, is built on open source developed with other NSIs, and the NSI contributes back. This keeps collective control over critical standardisation capabilities instead of depending on vendors.
SAF-P07	Implement loosely coupled processes and systems	Standardisation is independent from upstream acquisition and downstream editing. Standardized Data (State #2) is the decoupling interface: downstream consumers use standardised, classified data without knowing the original source structure.
SAF-P08	Data is shared property	Standardised data, once registered, is available to all authorised consumers. Standardisation is not repeated for each statistical product when the same standardised asset can serve several purposes.

Principle	Statement	Application in VS-02
SAF-P09	No data without metadata and classification	Standardisation is where classification is applied as a first-class activity. Every standardised dataset carries source-to-target mappings, classification versions, harmonised definitions, units, and a classification for availability, integrity, and confidentiality. The principle does not require raw inputs to arrive already classified; it requires that data handed over from State #2 onward carries the required metadata and classification. VS-02 is the principal point where this obligation is fulfilled.
SAF-P10	Security by design	Access controls, integrity verification, and audit logging are built into the standardisation pipeline from the start, so that every mapping and format-coercion operation is traceable and tamper-evident.
SAF-P11	Privacy by design	Direct identifiers are removed or pseudonymised during standardisation, at the transition from State #1 to State #2, applying data minimisation and purpose limitation by design instead of adding privacy after data is processed.

i Principles that apply across the process types

SAF-P09 is the defining principle of VS-02 and applies to all four process types: Design defines the target classifications and mappings, Execution applies and captures them, and Management assesses classification quality and closes remaining gaps. SAF-P11 applies to *Design Standardisation Registration and Privacy Framework*, to *Build Conformance Validation Pipelines and Metadata Registration*, and to VS-02.5, where the removal or pseudonymisation of direct identifiers is confirmed. SAF-P07 governs both the intake of State #1 from VS-01 and the handover of State #2 to VS-03.

8. Interfaces, services, and events

Boundary interfaces

VS-02 has two boundary interfaces: intake from VS-01, and handover to VS-03.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-01 → VS-02 (intake)	Registered raw data version (#1)	Minimum metadata set of State #1: provenance, source, collection method, receipt timestamp, file hash, agreement reference, classification	*Raw Data Registered*	VS-01 keeps custody of the raw artefact; VS-02 takes responsibility for standardisation on receipt of the event
VS-02 → VS-03 (handover)	Registered standardised data version (#2)	Minimum metadata set of State #2: mapping lineage, classification versions, recorded unit, currency, and reference-period bases, list of unmapped and uncoded values for VS-03 to code, definition-difference records, conformance assessment, pseudonymisation record, classification	*Standardized Data Registered*	VS-02 keeps custody of the standardised artefact; VS-03 takes responsibility for editing and enrichment, including any value conversion, interpretive coding, and reconciliation, on receipt of the event

Business services

Service	Offered to	Realised in	Interfaces
Standardisation service	Raw data consumers, standardisation teams	VS-02.1 Profile and Assess Data Structure; VS-02.2 Apply Standard Classifications and Formats	Event subscription (*Raw Data Registered*)
Classification mapping service	Standardisation teams, methodologists, other domains that need code mapping	VS-02.2 Apply Standard Classifications and Formats	Mapping interface from source code to standard code through deterministic correspondence, correspondence table lookup, code list exchange compliant with SDMX
Semantic harmonisation service	Metadata architects, downstream analysts	VS-02.3 Align Metadata and Semantics	Concept links and harmonised metadata in the catalogue
Conformance validation service	Process owners, quality managers	VS-02.4 Validate Structural and Semantic Conformance	Conformance assessment and audit trail

Service	Offered to	Realised in	Interfaces
Standardised data provisioning service	VS-03 Data Editing and Enrichment, data analysts, other domains	VS-02.5 Register Standardized Artefacts	Event notification (*Standardized Data Registered*) and a catalogue query interface to find and access standardised datasets and their lineage
Classification catalogue service	All data consumers in the NSI	Management sub-process *Manage Classification and Metadata Lifecycle*	Search and browse interface for classifications, code lists, and concept definitions, with metadata exchange compliant with SDMX
Design and methodology service	Methodologists, data architects, classification stewards	All Design sub-processes of VS-02	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artefacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Raw Data Registered	Start of VS-02.1 for that raw data version
Standardisation Conformed / *Standardisation Non-Conformant*	Outcome of VS-02.4; non-conformance starts re-mapping, routing of unmapped values to VS-03, or conditional acceptance
Standardized Data Registered	Completion of VS-02.5; triggers VS-03
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-02 event choreography between VS-01, NSI teams, and process management across Design, Implementation, Execution (raw data registered, profiling, one-to-one mapping and format coercion with unit and period metadata, conformance passed or failed, registration or re-mapping and routing to VS-03), and Management (key performance indicators, mapping quality, compliance, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-02	Receives from VS-02
VS-07 Governance and Compliance	Privacy and security policy, data and metadata governance	Audit trails of mapping lineage and pseudonymisation, privacy and standards compliance reports
VS-08 Innovation and Methodological Development	New mapping and classification techniques	Improvement needs; access to #2 for exploration under specific authorisation
VS-09 Infrastructure and Platform Enablement	Computing and orchestration for processing large datasets, storage, catalogue, event interfaces	Capacity and processing-window requirements

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
Mapping coverage	Share of source codes mapped one-to-one to a valid standard code	Per dataset, per classification
Unmapped-value rate	Share of values without a correspondence, routed to VS-03	Per source, per cycle
Conformance pass rate	Share of datasets passing conformance on the first run	Per source, per cycle
Format-conformance rate	Share of records conforming to the target structure and format	Per dataset, per cycle
Processing time	Time from raw data registration to standardised registration	Per dataset, per cycle
Re-mapping rate	Frequency of datasets returned for re-mapping or design change	Per source, per cycle

10. Examples

Labour Force Survey. The labour statistics team designs deterministic correspondence tables: source education codes map one-to-one to ISCED-2011, and source region codes map to NUTS-3. In each cycle the XML returns are coerced to the target schema and mapped through these tables. The 42,000 free-text occupation strings are not coded in VS-02. They are flagged and handed to VS-03, where occupation is coded to ISCO-08. No response value is altered; values are only re-expressed in standard codes where a deterministic correspondence exists.

Tax register intake. The quarterly corporate tax file, registered by VS-01, is profiled on arrival: 47 fields, the corporate identifier present at 99.8% coverage, economic activity supplied as the internal code of the tax authority, which has a maintained correspondence to NACE, and turnover in thousands of euro. The structure-gap assessment identifies the activity-code mapping and the format coercion of the XML as the VS-02 work. It records that the unit basis of turnover ("thousands of EUR") travels as metadata, and that conversion to a common unit is done in VS-03.

Multi-source business statistics. The business statistics data model uses the statistical business register identifier as the canonical linkage key, expresses economic activity in NACE Rev. 2 through a maintained correspondence table, and requires each turnover figure to carry its currency and reference-period basis as metadata. Survey returns, VAT register data, and structural business statistics are all aligned to this target schema. When turnover arrives from the structural business survey, the VAT register, and a commercial dataset, VS-02 links each turnover variable to the canonical concept and records that the VAT register reports on a fiscal-year basis and the survey on a calendar-year basis. The values are not adjusted to a common basis; the alignment is done in VS-03, so the difference stays transparent and the original values intact.

Web price data. A component coerces heterogeneous retailer price feeds into the canonical price record by reshaping fields and casting types. It records the currency, the unit (for example price per kilogram), and the scrape timestamp of each item as metadata, and stores prices exactly as received. Lineage links every standardised record to the exact raw scrape record. In conformance validation, all 11,875 records of a daily dataset conform structurally, every product whose source code has a COICOP correspondence is mapped, and the record count and observed price points match the raw scrape exactly. Products without such a code are flagged for VS-03, which also converts currencies and units to a common basis. The result is *Standardisation Conformed*, logged with rule version, engine version, and operator ID.

Toolkit: SAF-TK-V02 VS-02 maturity checklist and scenarios